Classify EEG and Reveal Latent Graph Structure with Spatio-Temporal Graph Convolutional Neural Network

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***Abstract*—Electroencephalogram(EEG) is a test that detect brain activities using multiple electrodes placed on the scalp. Multiple channels of EEG signals are recorded through the electrodes and are widely used in applications such as neurological disease diagnosis, emotion recognition, and behavior modeling. Recently, deep learning methods have been applied to classify EEG signals, where the different EEG channels are almost treated as a 2D grid input to the machine learning model. This data formation doesn't consider The complex connection among the EEG channels is not considered in such data formation. In our work, we treat EEG signals as frames of graph, and propose an end-to-end edge-aware spatio-temporal graph convolutional neural network for EEG classification. Specifically, we iteratively apply graph convolutional layer spatially and standard convolutional layer temporally. Since there is no prior knowledge about the exact connection among EEG channels, in our model, we initialize the connection as complete graph and apply learnable mask to capture graph structure at different levels. Furthermore, we also propose an iterative method based on information aggregation in graph convolution mechanism to reveal the latent graph structure. Empirical evaluation shows that our model achieves superior performance over state-of-the-art methods for EEG classification, and the learnt and revealed latent EEG graph structure is verified to be meaningful by neuroscientists.**

***Keywords*-EEG Classification, Graph Convolutional Neural Network, Spatio-Temporal Model, Deep Learning**

I. Introduction

Electroencephalogram (EEG) contains the most comprehensive information of human mental state. Thus, EEG is widely used for applications including emotion recognition, neurological disease diagnosis, and behavior modeling. Although multi-channel EEG signals can be sampled from electrodes to be classified for high-level tasks, the graph structure among the electrodes are rarely utilized. Besides, the correlation among the EEG signals is also not very explicit in neuroscience. But such correlation may contain rich insight. Thus, we aim to model EEG classification problem with the graph structure, and indicate the latent graph structure within the data-driven model.

Two facts motivate our work. The first is that, thanks to the release of EEG large datasets in recent years, neural networks is getting more attention with increasing power in the field of EEG classification. However, most previous research neglects or suffers from such problem [1]–[4]: it is difficult to capture the inconspicuous relationships among EEG signals from different channels. Such relationship naturally contribute significantly EEG analysis. Considering the latent correlation among EEG channels, we believe these channels are supposed to constitute a graph structure. The second fact is that recent studies of graph neural network (GNN) [5]–[8] show remarkable performance of GNN in kinds of applications [9]–[13]. We believe it is good time to try to build graph neural network model to classify such multi-channel EEG signals. With the power of GNN to well model graph structured data, it could be possible to reveal latent graph structure of EEG signals for certain task.

However, it is not easy to build such network and capture the latent graph structure. There are mainly three challenges. First one is how to represent the EEG signals. EEG is a kind of multi-channel time series data. We usually treat EEG as 2-D image when applying convolutional neural network, but the graph structure needs to be considered now. Secondly, the real latent graph structure is unknown. Although we know the spatial positional correlation among electrodes, such positional correlation is not functional correlation. We behave under the joint action of different parts of our brain, which means that the correlation between two positionally distant electrodes can be functionally close. Then, the third challenge is that even if we build the classification model successfully, the latent graph structure could be learned by layers of graph convolution. In another word, the graph structure captured by classification model could be dynamic, not static, which can be meaningless for neuroscientists.

To overcome above difficulties, we design an edge-aware spatio-temporal graph convolutional neural network and propose an information aggregation based method to indicate the latent graph structure. Specifically, we first treat EEG

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signals as frames of graph. We design certain blocks to iteratively apply graph convolution within the graph, by which we mean "spatially", and apply normal convolution among the graphs, by which we mean "temporally". Intuitively, at each time it comes to next moment, it should be helpful to reflect how the correlation among electrodes is now. Then, to step aside the problem of lacking known functional graph structure, we try to learn a weighted complete graph instead. We initialize latent graph as a complete graph and set a learnable matrix for each graph convolutional layer. Then, for each graph convolutional layer, we calculate the inner product of adjacency matrix of the complete graph and the learnable weight matrix of current layer, and feed such product to graph convolution together with EEG signal tensor. After training, we get a set of learned weight matrixes with dynamic graph information. With these weight matrixes, we propose to further indicate the latent graph structure. As we know, for a node in the graph, graph convolution operation means aggregating information from other nodes through edges and combine with the node [7], [14]. Besides, there is an intuition that the more important the edge is, the more information should flow through the edge. Based on the nature of graph convolution and the intuition above, we propose an information aggregation based method to extract latent static graph structure. Based on the mechanism of graph convolution operator and the learned weight matrix, we calculate the feature value flowing through the edge in the convolutional layers as the importance of the edge, and finally get an edge importance matrix to further reveal the latent graph structure. Our empirical evaluation includes model performance comparison with baselines and the latent graph structure indication for each time of cross validation. We ask two neuroscientists for feedback and receive great interest. The analysis and explanation of learned latent graph structure shows the effectiveness of our methods.

In summary, our contributions can be concluded as follows:

- We propose an edge-aware spatio-temporal graph convolutional network for EEG classification, where we treat EEG as frames of graph. Such model not only extracts spatial feature and temporal feature iteratively, but also learns the dynamic latent graph structure. Experiment shows its superior performance over other state-of-the-art methods.
- We propose an information aggregation based method with the learned weight matrix to reveal the static latent graph structure for EEG electrodes. Our visualization of latent graph indication shows interesting graph structure, which is confirmed further by two neuroscientists.

II. Related work

In this section, we mainly introduce related EEG classification methods and some kinds of interesting graph convolutional networks.

For EEG classification task, there are many feature engineering based methods [15]. For instance, [16] extracts features from time-frequency domain and wavelet transform domain and then feeds different classifier with the combination of the individual feature vectors. Also, many other features are investigated including features of wavelet entropy [17], features of wavelet coefficients [18]. Although these methods can relatively work beyond data limitation with domain knowledge, they are time consuming to engineer many features and strongly rely on knowledge and expert experience. To overcome this problem, more and more end-to-end deep learning models are developed. [19]–[21] utilize layers of 2-D convolution network for different applications such as EEG-based brain-computer interface and computer-aided EEG diagnosis. [22] utilize domain knowledge of cerebral hemispheric asymmetry to design an asymmetric LSTM network for EEG emotion recognition. These methods achieve good performance in certain task, but they neglect the graphical correlation among EEG channels. [23] attempts to incorporate such correlation by converting the chain-like EEG sequences into 2-D mesh-like hierarchy and apply both cascade and parallel convolutional recurrent neural network. Though step further, this paper is still limited by the conventional convolution which is not sufficient for graph structure.

To deal with data in more complex structure, graph neural network is getting pretty popular in recent years. Many remarkable papers are worthy to refer, such as graph attention network [24], In the graph neural network area, the most related work is spatio-temporal graph convolutional neural network [25]–[27]. [28] develop a scalable method for casting and arbitrary spatio-temporal graph as a rich RNN mixture. [25] and [26] propose two kinds of spatio-temporal graph convolutional networks to predict traffic situation with encoder-decoder structure. [27] proposes a new spatial temporal graph convolutional network which can automatically learn both the spatial and temporal patterns from data. In all of these work about spatial temporal graph convolutional neural network, there is explicit graph structure, which makes it not too difficult to build graph convolutional layers for the problem. In our EEG classification problem, the data is in graph nature but the real graph structure is latent. We manage to learn the graph structure and train our model in the meantime, and extract an edge importance matrix based on the nature of graph convolution to reveal the real latent graph structure.

As far as we know, there are only 2 papers also utilize graph convolutional network to classify EEG signal [29], [30]. [29] also design a dynamic graph convolutional neural network to learn the intrinsic relationship between different electrode under the task of emotion recognition. However, in [29], the input to neural network is not raw data but extracted features, and the learned graph is dynamic, consisting of several graphs, which is not investigated further. Not to

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learn graph structure, [30] chooses to define rules to build adjacency, and so the graph convolutional neural network can incorporate domain knowledge. But in the meantime, the performance is also limited by the rule defined. And none of both papers try to reveal the real latent graph structure.

III. Method

A. Classification Problem Formulation

Most of EEG related tasks can be formulated as classification problem, including EEG-based behaviour modeling, emotion recognition, abnormal detection and so on. We take one EEG signal as $S = [s_1, s_2, ..., s_n]$, consisting of $s_i = [x_1, x_2, ..., x_t]$, where *n* is the number of channels from different electrodes and *t* is the number of signal length. We take S as input and output one-hot class label $Y = [y_1, y_2, ..., y_m]$ with *m* as the number of classes. Then, we use categorical cross entropy loss for EEG classification:

$$L(S, Y) = \frac{1}{M} \sum_{i=0}^{M} logp(Y = y_i|S) \tag{1}$$

To produce the outcome *Y*, we build a deep neural network based on spatio-temporal graph convolutional block (ST-GConv), consisting of spatial graph convolutional layer (SG-Conv) and temporal convolutional layer (TConv). We believe there is a latent functional graph structure among *n* channels. Thus, we treat EEG signal *S* as *t* frames of graphs, and apply SGConv spatially on the graph. We use the basic graph convolutional layer described as [6] with formulation as follows:

$$GConv(f_{in}, A) = \sigma(D^{-\frac{1}{2}} A D^{-\frac{1}{2}} f_{in} W) \tag{2}$$

where $\sigma(.)$ denotes an activation function, $D = \Sigma_j A_{ij}$, f_{in} denotes input feature, A denotes the adjacency matrix of the graph. The key problem here, different from other tasks solved with spatial temporal graph neural network, is that the functional graph structure, as a main component of the graph convolution, is unknown. To sidestep such problem, we utilize complete graph and learnable mask to learn the graph structure when applying the graph convolution operator:

$$GConv(f_{in}) = \sigma(D^{-\frac{1}{2}} \ (W_\alpha \cdot \overline{A}) D^{-\frac{1}{2}} f_{in} W) \tag{3}$$

where $\overline{A}$ denotes the adjacency matrix of the complete graph and W_α denotes the learnable weighing matrix. In the meantime, to capture the pattern among these frames, we also apply normal convolutional layer temporally.

B. Model Architecture

In previous work, many methods treat multi-channel EEG signals as 2-D image and directly apply 2-D convolutional layer, shown as Figure.1. Such treatment neglects the latent graph structure among the channels. We believe the information in the latent graph structure can potentially improve EEG classification performance. Thus, we transform the EEG input into the format of frames of graph, where each node indicates a electrode. As for the edge of such graph, there is no prior knowledge to predefine a proper graph. Although there is spatial positional correlation among these electrodes which can be used to construct a graph, what we want to figure out is a latent functional graph, which means an edge should describe the correlation between the function of different part of brain for classification task, not between spatial position of electrodes. To step aside such problem, we utilize the adjacency matrix of complete graph and learnable weight matrixes to learn the latent graph structure during training.

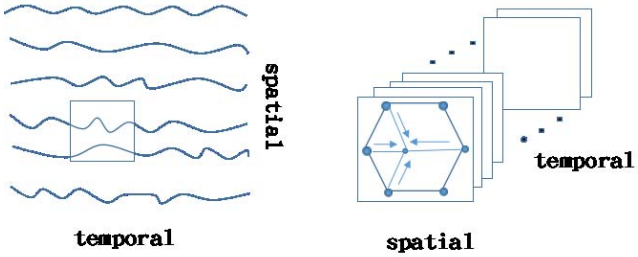


Figure 1. Input Format

Figure.2 shows our neural network model for EEG classification. We build whole structure based on spatio-temporal graph convolutional block (ST-GConv). We stack ST-GConv blocks and apply skip connection [31] between the input and output of ST-GConv block. Such ST-GConv block consist of two sub-blocks: graph convolutional block(GConv) and temporal convolutional block(TConv). Let's suppose the input tensor is in the shape of (*C*, *V*, *T*), where *C* denotes the number of channels, *V*, as "Vertice", denotes spatial dimension, and *T*, as "Temporal", denotes temporal dimension. The first dimension together with an adjacency matrix make up a graph structure. We design GConv block to apply graph convolutional operator within the graph of electrodes and their correlation, by which we mean "spatial", and TConv block to apply normal convolutional operator among different frames of the graph, by which we mean "Temporal". As for the input of GConv block, besides EEG tensor, we also feed GConv with the inner product of A and a learnable weight matrix, where A is the adjacency matrix of the complete graph for all electrodes as nodes, where the weight matrix is a learnable edge-aware mask. Thus, after epoches of training, result of such inner product can be regard as the adjacency matrix of a weighted directed digraph. After layers of ST-GConv blocks, we apply an FCN layer, a AveragePooling layer and a Softmax layer to get output and calculate loss. FCN layer is a standard convolutional layer with filters number of 1 and kernel size of 1 to map tensor into 1 channel, and the MaxPooling layer here is applied spatially and temporally in the meantime.

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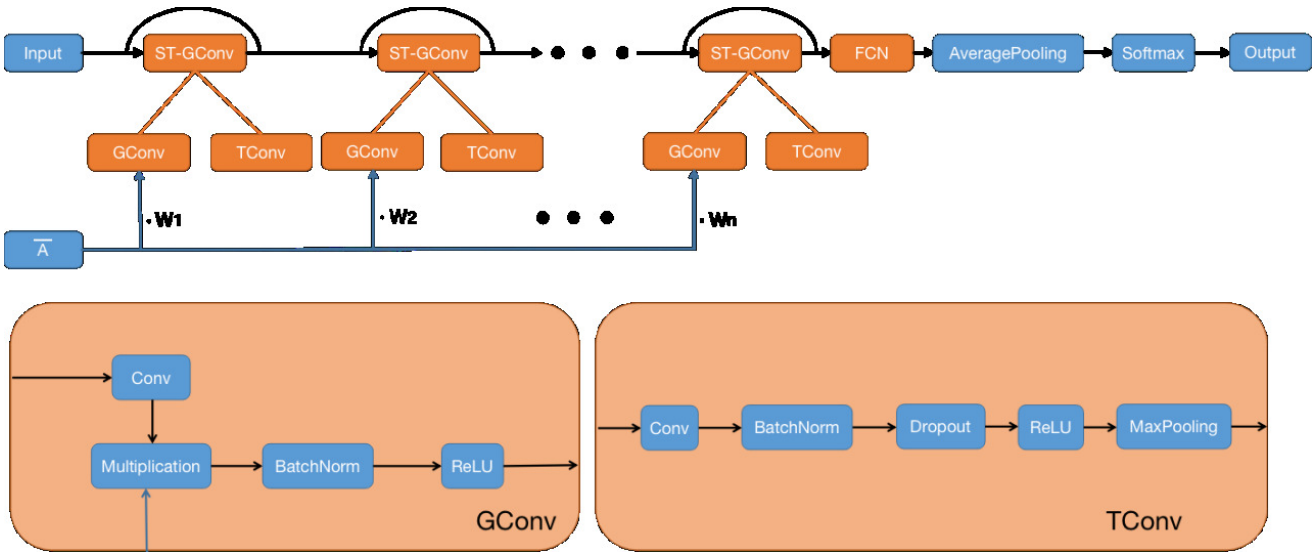


Figure 2. Model Architecture. Each ST-GConv block consists of a GConv block and a TConv block. GConv block is applied spatially to extract features among electrodes inside the graph, and TConv block is applied temporally to extract features among frames of graph. To feed GConv with adjacency matrix of the latent graph structure, we combine the adjacency matrix of a complete graph with a learnable weight in the same shape using inner product.

By stacking ST-GConv block, we iteratively apply GConv block and TConv block and extract spatial and temporal features in the meantime. Specifically, GConv block consists of 4 units. Input tensor first comes to a convolutional layer applied for the first dimension, and flows into a multiplication operator with another input of adjacency matrix. This convolutional layer and multiplication operator together implement the graph convolution in [6]. After the multipliation operator, there are a batch normalization layer [32], [33] and a ReLU activation layer. Similarily, in TConv block, there are a standard convolutional layer, a batch normalization layer, a dropout layer [34], a ReLU activation layer and a MaxPooling layer. The conlutional layer here is applied for the second dimension, to compress information temporally, and the MaxPooling layer here is applied only temporally. For each ST-GConv block, there is an optional skip connection between the input and output of the ST-GConv block. If the channel numbers of such input and output are not the same, we further add a convolutional layer with kernel size of 1 in the skip connection, to transform the input into a tensor with the same number of channels as output. Then, we add them together.

C. Latent Graph Structure Indication

In the above model structure, we step aside the problem of lacking prior graph structure with the complete graph and learnable weighing matrixes. Then, after training, we can get a set of adjacency matrixes, in another word, a set of graph structures, at different abstract level. Since we optimize the parameters of such matrixes and all other parameters in the model at the same time, the latent graph structure of electrodes and discriminative patterns are encoded together. Thus, although we can get dynamic graph structures, the real static latent graph structure is lost during the training process. However, we still hope to capture the real static graph structure because such graph structure is very worthwihle for medical study and for further possible performance improvement.

To further reveal the latent graph structure, we propose a method to calculate edge importance matrix with information aggregation between graph convolutional layers. Our method is based on two points. One is an important nature for graph convolutional layer: each node is aggregating information from other nodes through the edges. The other is the intuition that the more important the edge is, the more feature value should flow through the edge. Thus, based on such two points, we can calculate absolute value flowing through each possible edge for graph convolutional layers to get the edge importance matrix for each graph convolutional layer, and average these matrix to get the edge importance matrix for the classification task, which reveals the static latent graph structure.

Information aggregation is illustrated as shown in Figure.3. There is a graph at graph convolutional layer l on the left and a graph at graph convolutional layer $l+1$ on the right, with 4 nodes i, j, m and n. Let e_{ij} denote the edge from node i to node j. For node i in graph convolutional layer l, graph convolution operator aggregate information from node m, n, j respectively through edge e_{mi}, e_{ni}, e_{ji}. If the graph is weighted, the amount of information can be further multiplied with the corresponding weight in the weighted adjacency matrix. If we know how graph convolution combines

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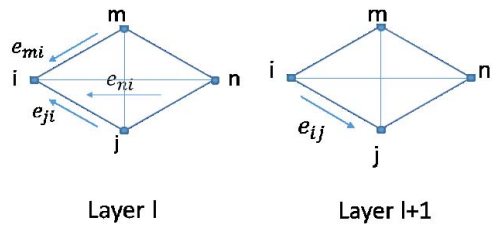


Figure 3. Information aggregation with respect to node *i* and edge e_{ij} between graph convolutional layer *l* and graph convolutional layer *l+1*. This figure shows how node *i* aggregates information in graph convolutional layer *l* and send information to node *j*, which is supposed in a complete and unweighted graph.

the node *i* and the information aggregated from its neighbor nodes, we can get the new feature value of node *i* after applied graph convolution. Then, in the next convolutional layer, we can calculate the value flowing from node *i* to other nodes if the weighted adjacency matrix for the next convolutional layer is known. Further, if all of the weighted adjacency matrixes are known, we can calculate the relative amount of information for each edge from the first layer to the last layer iteratively. In our setting, after training, both of graph convolution operator and the learned weighted adjacency matrix for each convolutional layer are known. Therefore we are able to get the final edge importance matrix.

Let Q^l denotes the edge importance matrix of the layer *l*, Q^l_{ij}) denotes the value flowing through the edge e_{ij} from node *i* to node *j* in the layer *l*, A^l denotes the adjacency matrix of the *l* graph convolutional layer, A^l_{ij} denotes the elements of the *i* row and *j* column in the A^l, and h^l_i indicates the feature of node *i* in the graph convolutional layer *l*. Then, we define the graph convolution operator as below:

$$h^l_i = ReLU(W \cdot MEAN\{h^{l-1}_j \cdot A^l_{ji}, \forall j \in \mathcal{N}(i) \bigcup \{i\}\}) \tag{4}$$

where $ReLU$ is the activation function, W is a learnbale matrix inside graph convolution operator, $MEAN$ is the element-wise *mean* pooling, $\mathcal{N}(i)$ indicates the neighbor nodes of node *i*. The graph convolution operator is an edge weighted version of [6]. Since all of the weighted matrix are known, given input tensor, we can iteratively calculate all node features in each convolutional layer. Then, to calculate the importance of edge e^l_{ij}, we have:

$$\begin{aligned} Q^l_{ij} &= A^l_{ij} \cdot h^l_i \\ Q &= \sum_{l=1}^{L} Q^l / L \end{aligned} \tag{5}$$

where L is the number of graph convolutional layer. So we can get Q^l for all graph convolutional layers, and average them to get Q which contains the information of the real static latent graph structure. To further get the undirected edge importance matrix, we add each Q_{ij} to Q_{ji}, normalize the Q and set each Q_{ii} as zero. Then, we get the final edge importance matrix which indicates the static latent graph structure.

After we get the final edge importance matrix, we further need to confirm its effectiveness. We train several models and extract the edge importance matrix for each model. We want to show whether the models respectively trained capture similar key features, and whether these feature can be confirmed according to neuroscience. Thus, we first design a web-based interactive tool to help user explore the graph constructed with our edge importance matrix. And then, we cooperate with two neuroscientists to check details of extracted graph structure. More details of our model training and graph structure analysis are described in next section.

IV. Experiments

In this section, we describe our experiment settings, results and analysis. We introduce details of dataset, and we choose two state-of-the-art model as baseline. Then, we explain the details of model, including training and validation setting. For the next, we also describe how we calculate and visualize the latent graph structure indication. Furthermore, we analyse the performance of models and the visualization of the latent graph structure indication.

A. Dataset

We use a public EEG Motor Movement/Imagery Dataset for experiments from PhysioNet [35]. This dataset consists of over 1500 one- and two-minute EEG recordings, obtained from 109 volunteers. EEG is of 64-channel, recorded using the BCI2000 system when subjects performing different motor/imagery tasks. We choose the dataset for the task of imagining opening and closing left or right fist. The detail of the task is described as below. The subject stand in front of a screen. When the task begins, a target appears on either the left or the right side of the screen. The subject imagines opening and closing the corresponding fist until the target disappears. Then the subject relaxes. The electrodes of EEG is shown as Figure.4 After training our model, we further show our model indicates how the edges among these electrodes looks like.

B. Baseline

We choose Deep ConvNet model and Shallow ConvNet model from [2] and [3]. Both models are specially designed with reference to previous successful methods. As described in [2], Deep ConvNet consists of four convolution-max-pooling blocks, with a special first block designed to handle

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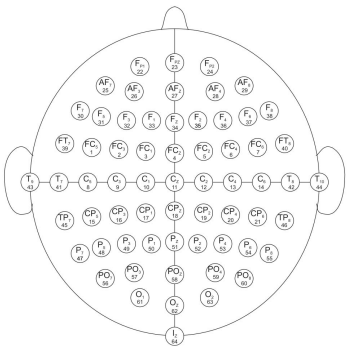


Figure 4. EEG Electrodes. This figure shows the electrodes in international 10-10 system.

input tensor, followed by three standard convolution-max-pooling blocks and a dense softmax classification layer. The first convolution block was split into a first convolution across time and a second convolution across electrodes. The Shallow ConvNet is in a relatively shallow structure, which is inspired by FBCSP. FBCSP is a widely-used method to decode EEG data and achieved the best performance for the BCI competition IV dataset 2a. Blocks of the Shallow ConvNet is are designed analogous to treatments in FBCSP. But unlike in FBCSP, all steps are jointly optimized in the Shallow ConvNet. Thus, the Shallow ConvNet outperform FBCSP statistically, and we take both the Deep ConvNet and the Shallow ConvNet as baseline.

C. Model Setting

Based on the model architecture in Figure.2, we stack 5 ST-GConv blocks. We respectively assign filters number of ST-Conv blocks with 32, 32, 64, 64, 128, and set kernel size of each ST-Conv block as 17, set dropout rate of dropout layer in each TConv block as 0.5, and set pooling size of max pooling layer in each TConv blocks as 2. We set branch size as 16 and utilize *Adam* optimizer with initial learning rate of 0.001. Considering that the shape of input tensor is (16, 64, 497, 1), where 64 is the number of electrodes and 497 is the number of temporal length, the output of FCN layer is in the shape of (16, 64, 16, 1), so we further apply AveragePooling layer with kernel size of (64, 16) on the second and third dimension. Then, the tensor flow through Softmax layer and output classification result.

As for training strategy, we split dataset as 8:1:1 into training set, validation set and test set. We apply 5-fold cross validation, and run each model for 5 times to calculate average result. For each fold in cross validation, we sample data from 80 subjects as training set, data from other 10 subjects as validation set, and data from another 10 subjects as test set. We also apply early stop with patience of 10 and learning rate schedule with patience of 5 and ratio of 0.1. We always remember the model of best performance during training.

Besides the two baseline model, to further validate our learnable graph setting, we add two additional models. The first one additional model remove the learnable matrixes and feed GConv blocks with an adjacency matrix of a graph without edges, which is a zero matrix. We mark such model as "ST-GCN-No-Graph". The other additional model also remove the learnable matrixes but feed GConv blocks with an adjacency matrix of a predefined graph. We predefine the graph by adding edges between nodes based on real spatial positional correlation of electrodes. The predefined graph structure is shown as Figure. 5. We mark such model as "ST-GCN-Predefined-Graph", and mark our model with learnable matrixes as "ST-GCN-Learnable-Graph".

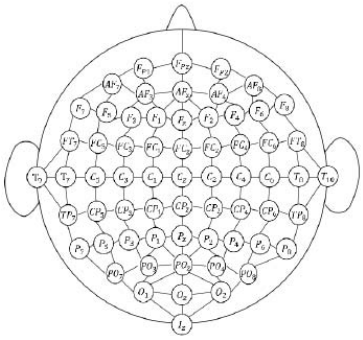


Figure 5. Predefined Graph Structure. Relative position of each node is the totally same as Figure.4, where a "node" indicates a electrode here. We connect nodes according to the real spatial positional correlation among these electrodes.

D. Calculation and Visualization of Latent Graph Structure

At every time training procedure stops, we record absolute value of the learnable weight matrix for each graph convolutional layer. In one time for 5-fold cross validation, for each fold, we calculate the edge importance matrix following the steps described in III-C with certain input tensor. Specifically, after training process, we reload the model, extract the learned weight matrixes for graph convolutional layers. Then, according to III-C, since the weight matrixes are got, which means the learned adjacency matrixes are got, the next key step is to get the feature of nodes h^l for each layer. So, we first randomly extract a batch of EEG signals from test set as input. Then, instead of extracting weights and doing calculation layer by layer, we feed the input to model and extract the output tensor of node features from each graph convolutional layer. Thus, since we have got the adjacency matrixes and node feature for each layer, we can get the final edge importance matrix, for each fold. Then we average the 5 edge importance matrixes for the 5-fold cross validation. At last, we get 5 edge importance matrixes respectively for 5 times of 5-fold cross validation.

Then, we design a visual representation and web-based tool to show top-k edges with largest weight, where number

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k is defined by user. We visually encode the actual spatial position of each electrode into the coordinate in the web page. Further, to imply relative importance among the top-k edges, we also encode importance of the edge into the opacity of the corresponding red line. Based on such tool, we cooperate with two neuroscientists from the hospital affiliated to our university, to dive into the edge importance matrixes and investigate the static latent graph structure among electrodes.

E. Performance Result

We run 5 times of 5-fold cross validation for each model, and calculate the average area under the curve(AUC). The result is shown as Table.I.

Table I
PERFORMANCE COMPARISON

Model	AUC
Deep ConvNet [2] [3]	0.8071
Shallow ConvNet [2] [3]	0.7967
ST-GCN-No-Graph	0.7904
ST-GCN-Predefined-Graph	0.8140
ST-GCN-Learnable-Graph	0.8378

Compared to the two baseline models, ST-GCN-No-Graph model achieves low AUC. Since ST-GCN-No-Graph model has lost the graph structure, its GConv block does not work, and lost graphic information. Then, ST-GCN-No-Graph model naturally perform worse than the two baseline models.

In contrast to the ST-GCN-No-Graph model, ST-GCN-Predefined-Graph model utilizes a predefined graph structure based on spatial positional correlation among electrodes. As we know, signals from positionally close electrodes are usually similar because of the electricity streaming. Such positional prior knowledge should somehow help model perform classification better. As we can see, although the margin of performance improvement is not large, ST-GCN-Predefined-Graph achieve a little better performance than the two baseline models.

However, we still believe the predefined graph is not a good solution. For example, when we classify that a person is clapping or running, the best way is not to decide by measuring movement of two key points only on one hand, but to decide by measuring relative movement of two key points on each hand. Thus, we hope to capture the real functional correlation among electrodes and apply learnbale weight matrix on complete graph. As we expected, ST-GCN-Learnable-Graph achieves the best performance of these models, 2 points more than ST-GCN-Predefinede-Model and 3 points more than the two baseline models.

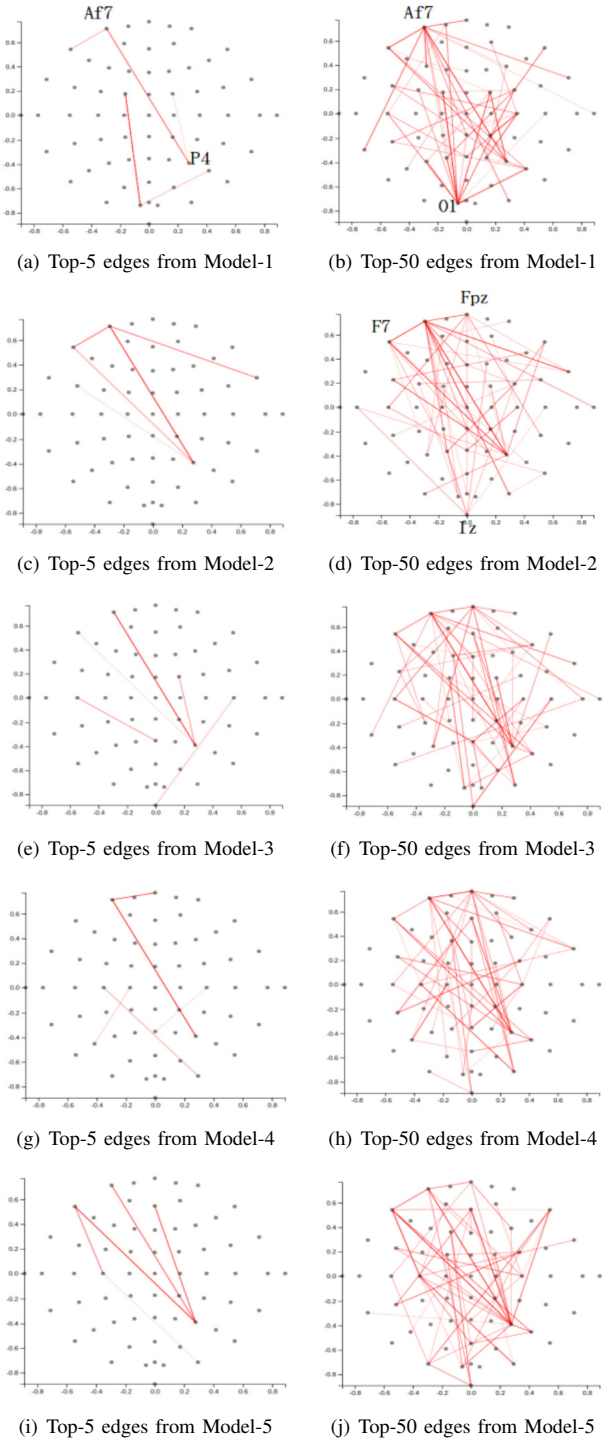
F. Latent Graph Structure Indication

Although our ST-GCN-Learnable-Graph model achieve superior performance, what ST-GCN-Learnable-Graph model has learned by the learnable weight matrixes is actually a dynamic graph structure with several adjacency matrixes. It is because that part of the information of the latent static graph structure is encoded in other parameters in the model during training procedure. So, we propose an information aggregation based method to calculate the edge importance matrix for the latent static graph. We calculate and visualize such matrix as described IV-D. Then, in this subsection, we further investigate what the edge importance matrix can indicate and whether the structures shown in the edge importance matrix can match medical concept.

To do such investigation, we cooperate with two neuroscientists from the hospital affiliated to our university. We share and introduce a web-based interactive tool to show top-k edges in the visualization of the international 10-10 system. After generally expressing what we have done, we also describe the details of the classification task since the task procedure can be important for explaining the latent graph structure. After some time to explore the 5 edge importance matrixes with our tool, they find some interesting insight. We share their analysis and explanation as below.

At last, they choose top-5 edges and top-50 edges to visualize as shown in Figure.6. Graphs in the same row come from the same edge importance matrix. One obvious interesting appearance is that all of the the top-5 edges from every edge importance matrix contain the same edge from electrode “Af7” to electrode “P4”, and the weights of this edge are all very high, indicated by the visual encoding of weight value to red color. In neuroscience [36], electrode “Af7” is in the region of prefrontal lobe, and the prefrontal lobe [37] is responsible for memory, emotion, language and movement. Electrode “P4” is in the region of parietal lobe [38], and the parietal lobe is mainly responsible for high-level sense including somatosensation, and sensation of spatial position. Recall what our classification task is: after seeing the target on the right or left side of the screen, the subject imagines opening and closing the corresponding fist until the target disappears. This, they explain as follows. When the target shows up, the subject automatically notices and memorize the target. At this time, prefrontal lobe participates in the procedure and causes neuroexcitation. Next, subject start imagining to act. In this procedure, the subject does not actually move, but only imagines, which needs the sense of movement. Such sense of movement is exactly what parietal lobe is responsible for, and the electrode “P4” located in parietal lobe. Furthermore, there are actually many nerve fibers connecting prefrontal cortex, premotor cortex and posterior parietal cortex, and the neuroexcitation is conducted from the prefrontal lobe to parietal lobe when the subject starts imagine, which is from where the electrode “Af7” is to where the electrode “P4” is. Such procedure could be the same for most subjects, and so our model learns such noteworthy feature and transforms much information between the two nodes corresponding to the electrode “Af7”

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(a) Top-5 edges from Model-1 (b) Top-50 edges from Model-1

(c) Top-5 edges from Model-2 (d) Top-50 edges from Model-2

(e) Top-5 edges from Model-3 (f) Top-50 edges from Model-3

(g) Top-5 edges from Model-4 (h) Top-50 edges from Model-4

(i) Top-5 edges from Model-5 (j) Top-50 edges from Model-5

Figure 6. Latent graph structure indication from 5 edge importance matries for each row.

and “P4”. As for in the top-50 edges, such appearance is also pretty conspicuous. Besides the electrode “Af7”, there are also many edges connected to the electrode “F7” and the electrode “Fpz”. These three electrodes are all in the region of prefrontal lobe, where is pretty active.

Besides the 4 mentioned electrodes, there are only 2 electrodes usually receive many edges, the electrode “O1” and the electrode “Iz”. What is interesting is that the both electrodes are in the region of occipital lobe, and occipital lobe is where visual center is. As we know, in the task, the subject sees the target and then images, which means visual function is also important in such procedure. Then, we can conclude that three importance factories: “vision” corresponding to the electrodes “O1” and “Iz”, “memory” corresponding to the electrodes “F7”, “Af7”, “Fpz”, and “somatosensation” corresponding to the electrodes “P4”, all can be confirmed by our edge importance matrix.

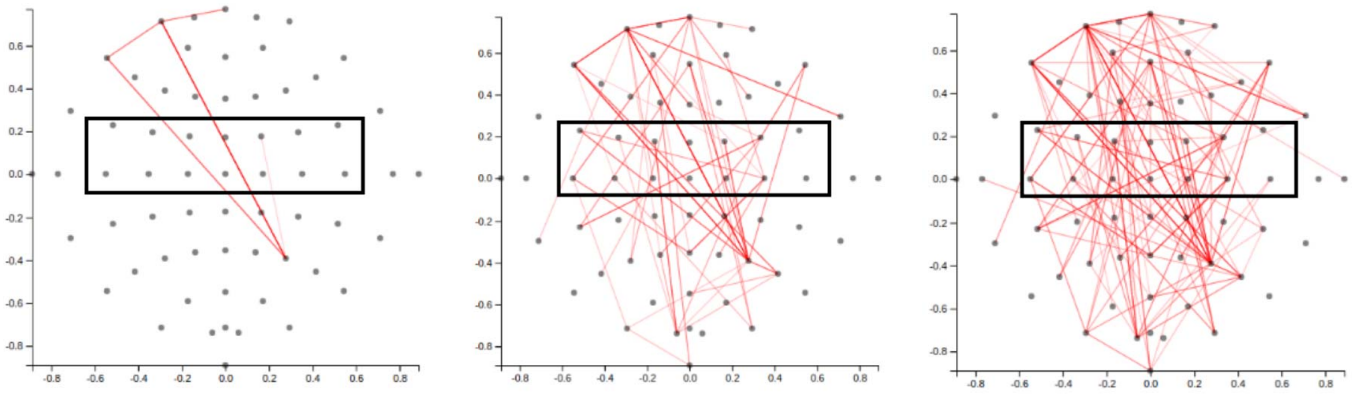
Furthermore, since the task is to imagine action, not to really act, it is natural to ask such question: how does the region with function to control movement behavior in the latent graph? To show statistical result better, we further provide the two neuroscientists with the average matrix of the 5 edge weight matrixes. With the average matrix, they pick out 3 figures as shown in Figure.7. In neuroscience, human control movement mainly with neurons in precentral gyrus, which is in the area of electrodes “FC”s (Fz, F1, F2, ..., F8) and “C”s (Cz, C1, C2, ..., C6). We mark this region with black rectangle in the figure. Although some crossing edges overlap this region, we can still find that the edges connected to the electrodes in this rectangle is relatively much less than the 6 electrodes we mentioned above.

Through above three facts, it can be confirmed that our model captures the main feature of the task, and the edge importance matrix reveals interesting latent graph structure.

V. Conclusion

In this article, we propose an edge-aware spatio-temporal graph convolutional neural network for EEG classification. We take input EEG as frames for graph, and apply graph convolutional blocks and temporal convolutional blocks iteratively to extract spatial feature and temporal feature in the meantime. To step aside the problem of lacking known functional graph structure, we apply learnable mask to automatically learn the graph structure, feeding each graph convolution with the inner product of the adjacency matrix of a complete graph and a learnable weight matrix. During the training process, the weight matrixes learn the dynamic latent graph structure. Then, we propose an iterative method based on the mechanism of graph convolution and intuitive observation to calculate edge importance matrix with the weight matrixes, which indicates the static latent graph structure for further research. Empirical evaluation shows superior performance of our model over state-of-the-art models for EEG classification. Besides, the edge

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(a) Top-5 edges (b) Top-50 edges (c) Top-100 edges

Figure 7. Latent graph structure after averaging the 5 edge importance matrixes. The black rectangles indicate the region to control body movement at the center of the graphs.

importance matrix indicates interesting latent graph structure matching neurological activity, with neuroscientists' confirmation. Future work includes investigating methods to improve classification performance based on the extracted latent graph structure and trying on other domain and tasks.

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